

HW-06 — Output Analysis: Warmup, Replications, and CRN

Module M08 | Chapters 10–11 | Weight 10% | Difficulty **

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Overview

Apply output analysis rigor to an M/M/1 queue: diagnose transient bias, determine the warmup period, plan the number of replications for a target CI half-width, and use Common Random Numbers to compare two configurations.

1 System

M/M/1 queue: $\lambda = 0.8/\text{min}$, $\mu = 1.0/\text{min}$, $\rho = 0.8$.

Theoretical steady-state:

$$W_q^* = \frac{\rho}{\mu - \lambda} = 4.0 \text{ min}, \quad W^* = 5.0 \text{ min}, \quad L = 4.0, \quad L_q = 3.2$$

2 Parts

2.1 Part A — Transient Bias and Warmup Detection (30 pts)

1. Run 10 replications of 2,000 customers each (empty-and-idle start). Plot the *cumulative sample mean* $\bar{W}_q^{(i)}(n)$ for each replication as a function of customer index n .
2. Apply Welch's moving-average method: average the 10 curves, smooth with a window of 40 observations, and plot the smoothed curve alongside the theoretical $W_q^* = 4.0 \text{ min}$.
3. Choose a warmup deletion point ℓ^* (in customer-index units) and justify your choice. What is ℓ^* converted to simulation-time minutes?
4. Demonstrate the bias effect: report the mean W_q computed (a) from customer 1, and (b) from customer $\ell^* + 1$ onward. How much does the bias inflate the estimate?

2.2 Part B — Sequential Replication Planning (25 pts)

1. Run a **pilot study** of $n_0 = 10$ replications (each 5,000 customers, deleting ℓ^* warmup). Record $\bar{W}_q$ and sample variance S^2 .

2. Target half-width $h^* = 0.25$ min at 95% confidence. Using the sequential formula $n \geq (t_{\alpha/2, n_0-1} \cdot S/h^*)^2$, compute the required total replications n^* .
3. Run the remaining $n^* - n_0$ replications and report the final CI. Verify that the achieved half-width $\leq h^*$.

2.3 Part C — CRN Paired Comparison (25 pts)

Compare two configurations:

- **System A:** M/M/1, $\lambda = 0.8$, $\mu = 1.0$ (baseline)
 - **System B:** M/M/2, $\lambda = 0.8$, $\mu = 0.7/\text{server}$ (same offered load $\rho = 0.8/0.7 \approx 0.57$ per server)
1. Run 30 **independent** replications of each; compute 95% CI for the difference $\Delta = W_q^A - W_q^B$.
 2. Run 30 **CRN paired** replications (same seed for both systems in each pair). Compute $\bar{D}$, S_D , and the 95% CI for $E[\Delta]$.
 3. Report the variance reduction ratio $S_D^2 / (S_A^2 + S_B^2)$. Explain in 2 sentences why sharing the arrival stream induces positive correlation between the two systems.

2.4 Part D — Conceptual Reflection (20 pts)

Answer the following in 2–4 sentences each:

1. Why does starting a steady-state simulation empty-and-idle bias results upward?
2. A colleague argues: “I ran one long simulation instead of 30 short ones; I have more data, so my estimate is better.” Where is the flaw?
3. CRN reduced variance in Part C. Describe a situation where CRN would *not* help (or could hurt).

Grading Summary

Component	Points
Part A — Warmup / Welch plot + ℓ^* + bias demo	30
Part B — Sequential pilot $\rightarrow$ target half-width	25
Part C — CRN comparison + variance reduction ratio	25
Part D — Conceptual reflection (3 questions)	20
Total	100

Full rubric: [rubrics/homework_rubric.pdf](#) | Solutions: the instructor package (available to instructors on request).